

JACOB PFAU

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EDUCATION

New York University, Bowman Lab PhD Student NYU Alignment Research Group, Center for Data Science Projects include: Evaluating generalization of RLHF fine-tuning in language models: truthfulness, mode-collapse and corrigibility. Modeling transformer use of adaptive compute in inference.	Fall 2022 - Present
University of Edinburgh MSc Mind, Language and Embodied Cognition (Philosophy Department). First class honors Dissertation: 'A decision theoretic perspective on bluffing in games of perfect information'	Sept. 2021 - Aug. 2022
Ecole Polytechnique (Université Paris-Saclay), Paris M1, Masters Computer Science, Data Science track. GPA 3.91	Sept. 2018 - Aug. 2019
Amherst College BA Mathematics & 5 College Logic Certificate. GPA (Math/CS/Logic) 3.73, Overall GPA 3.64	Sept. 2013 - May 2017
ETH Zurich Visiting student, Mathematics. GPA 5.34/6.0	Jan. 2016 - Aug. 2016

EMPLOYMENT

University of California, San Francisco. Keiser and Wei Labs <i>Bold and Basic Fellow</i> - Developed interpretability methods and benchmarking datasets leading to multiple publications. Mentored rotating PhD and medical students - Developed automated diagnosis model and semantic segmentation model for localizing skin lesions. Improved bias and uncertainty calibration.	Mar. 2019 - Aug. 2020
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PUBLICATIONS (BY RELEVANCE)

Objective Robustness in Deep Reinforcement Learning. https://arxiv.org/abs/2105.14111 Koch J, Langosco L, Pfau J, Le J, Sharkey L. ICML (2022)
Robust Semantic Interpretability: Revisiting Concept Activation Vectors. https://arxiv.org/abs/2104.02768 Pfau J, Young AT, Wei J, Wei ML, Keiser MJ. ICML Workshop on Human Interpretability (2020)
Global Saliency: Aggregating Saliency Maps to Assess Dataset Artefact Bias. https://arxiv.org/abs/1910.07604 Pfau J, Young AT, Wei ML, Keiser MJ. NeurIPS ML4Health Workshop (2019)
Stress Testing Reveals Gaps in Clinic Readiness of Image-Based Diagnostic Artificial Intelligence Models Young AT, Fernandez K, Pfau J, et al. npj Digital Medicine (2020)
Artificial Intelligence in Dermatology: A Primer. https://doi.org/10.1016/j.jid.2020.02.026 Young AT, Xiong M, Pfau J, Keiser MJ, Wei ML. Journal of Investigative Dermatology (2020)
Artificial Intelligence in Teledermatology. https://doi.org/10.1007/s13671-019-0259-8 Xiong M, Pfau, J, Young AT, Wei ML. Curr Derm Rep (2019)

FELLOWSHIPS AND AWARDS

Bold and Basic Fellowship, University of California, San Francisco Awarded competitive yearlong research fellowship for project using deep learning to connect skin cancer image data with vectorized genomic data	Sept. 2019 - Aug. 2020
Research Internship Prize, Ecole Polytechnique, Paris One of three masters students awarded excellence in research prize. For explainable deep learning work done at UCSF	Oct. 2019
Labex DigiCosme Fellowship, Labex DigiCosme, Paris Full scholarship for tuition and cost of living of masters at Ecole Polytechnique	Sept. 2018 - Mar. 2019

SKILLS AND MISCELLANEOUS

LANGUAGES: English (native), German (fluent), French (conversant), Mandarin (basic)

GRE SCORES: Overall 99%, Verbal 170 (99%), Quantitative 167 (92%), Writing 5.5 (98%)